# Heat Flow Across a Material Interface

## Conservative FDM and piecewise-linear FEM

MTH321 Project 2

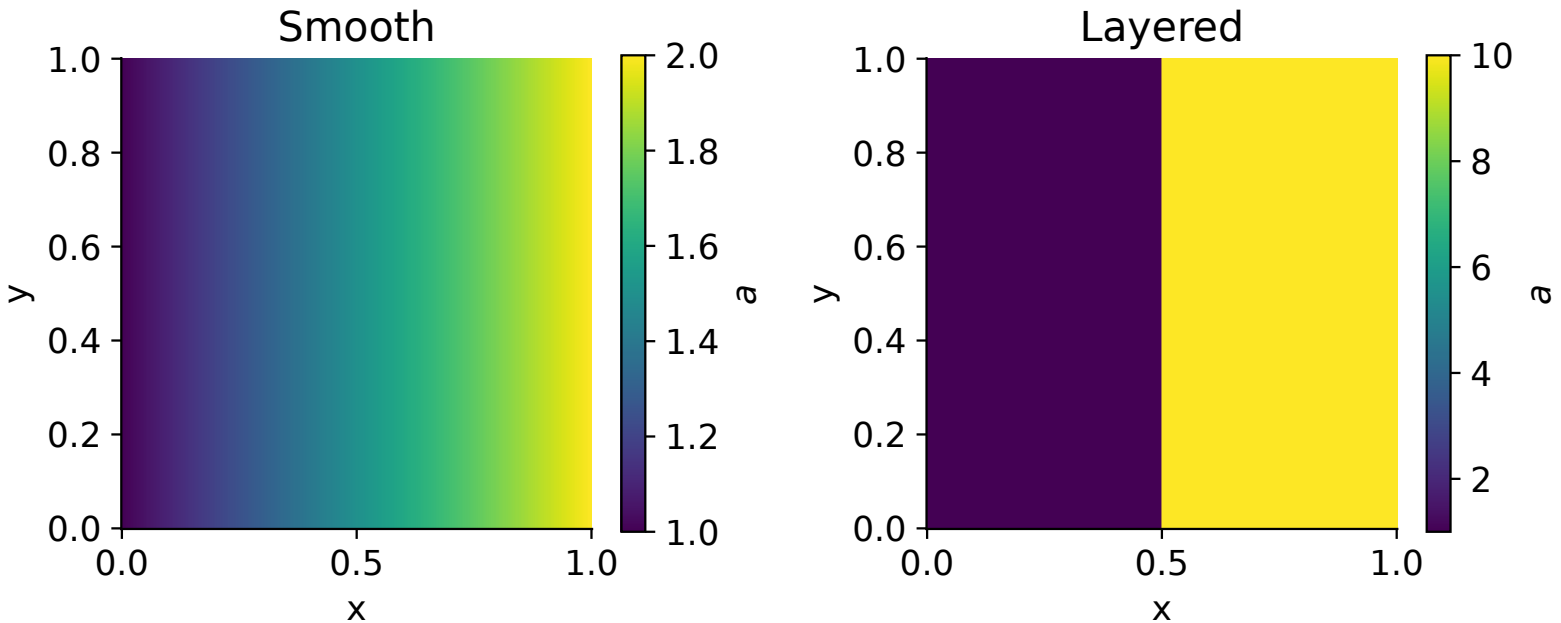
XJTLU

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What must a numerical method preserve when conductivity jumps?

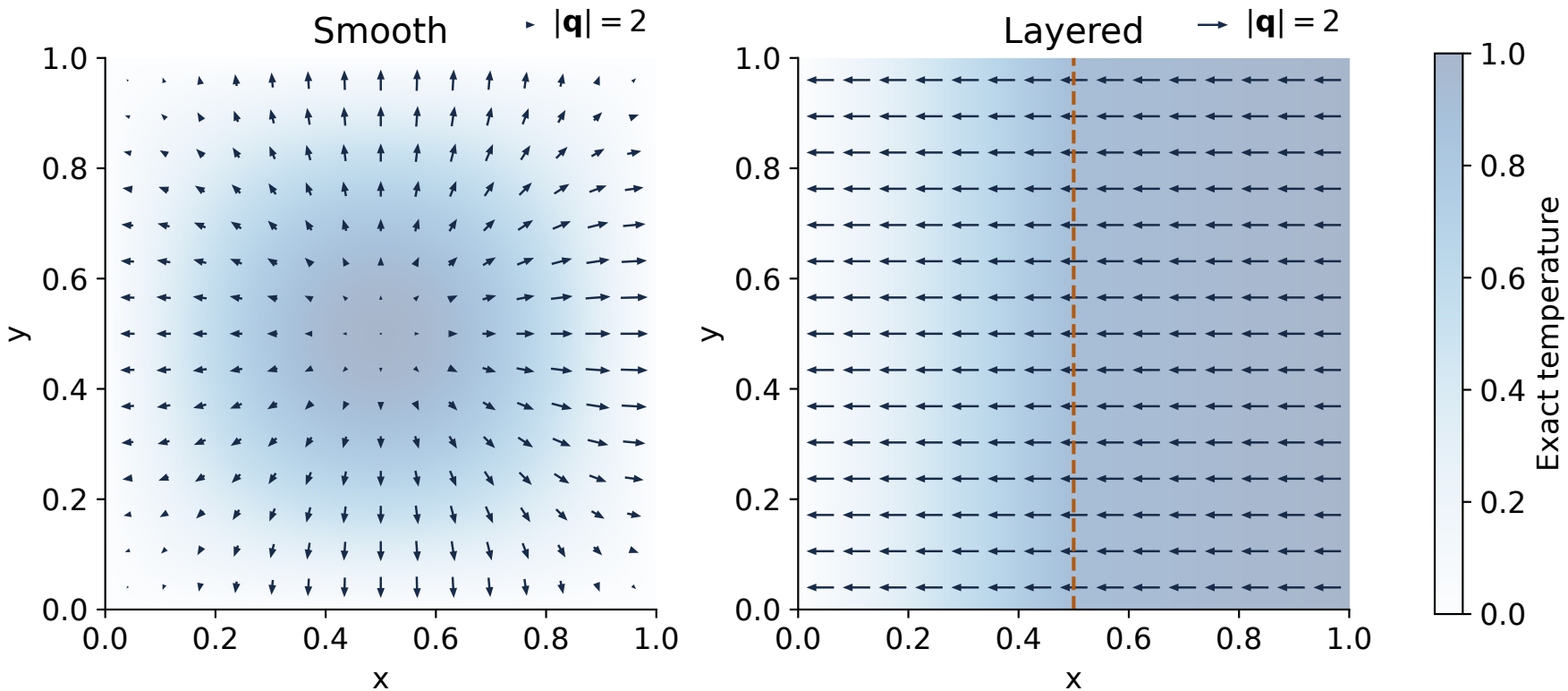
# Two conductivity models

$$-\nabla \cdot (a \nabla u) = f, \quad \mathbf{q} = -a \nabla u, \quad u = g \text{ on } \partial\Omega.$$




On $\Omega = (0,1)^2$: smooth $a = 1 + x$ tests accuracy; layered $a_L = 1$, $a_R = 10$ tests interface physics. Colors show conductivity; the two color scales differ.

# Heat flux in the two models




Background: exact temperature. Arrows: exact $\mathbf{q} = -a\nabla u$. Each panel has its own arrow key; compare magnitudes through those keys.

The layered flux points left and is constant across the interface $x = 1/2$.

# What the interface must preserve

For perfect contact with no interface source,

$$[u] = 0, \qquad [\mathbf{q} \cdot \mathbf{n}] = 0.$$

The layered benchmark has $f = 0$ and exact solution

$$u^*(x) = \begin{cases} 20x/11, & x \le 1/2, \\ (2x+9)/11, & x \ge 1/2. \end{cases}$$

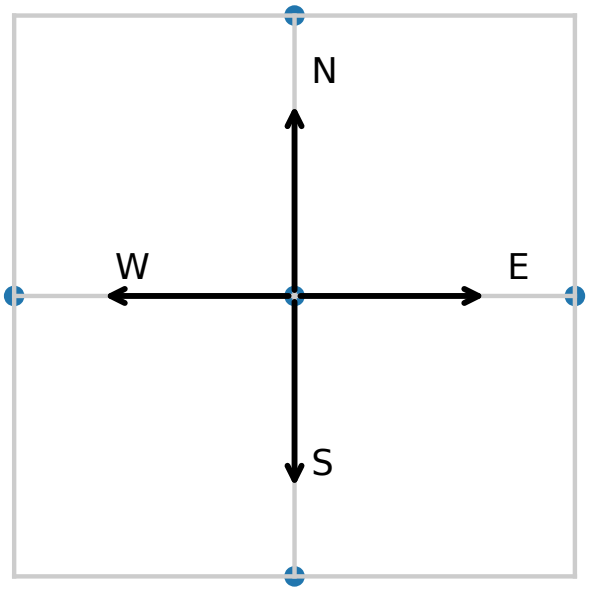
$$u_x^- = 20/11, \quad u_x^+ = 2/11, \quad q_x^* = -20/11.$$

Temperature is continuous; its slope changes so that flux stays continuous.

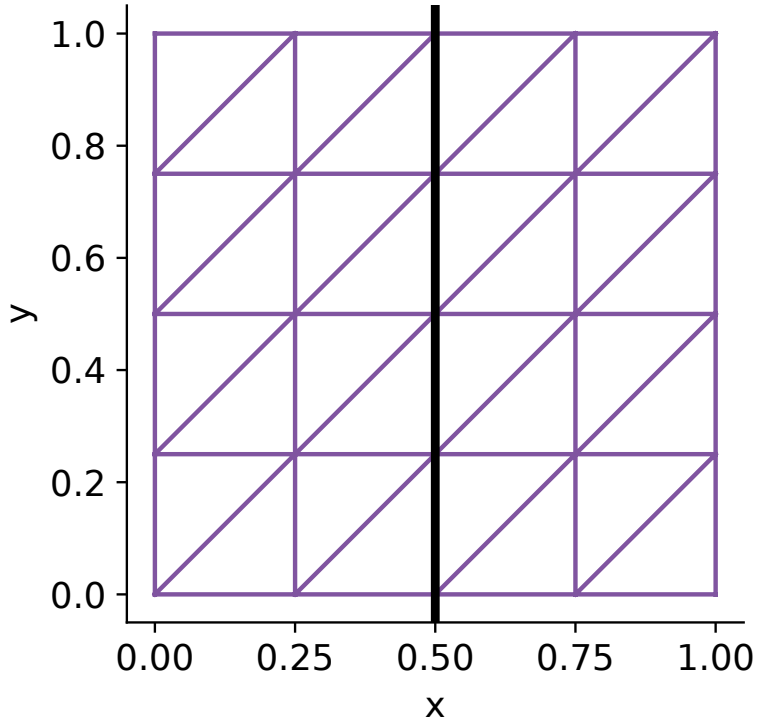
We impose $g = u^*$ on **all four boundary sides**.

# Two numerical representations

FDM: shared face fluxes




FEM: interface-fitted triangles




**FDM**: balance fluxes shared by neighboring nodes.
**FEM**: approximate temperature by continuous affine functions on triangles.

Left arrows indicate stencil directions. The black line on the right is the fitted material interface.

# FDM: coefficients and boundary assembly

$$q_{i+1/2,j} = -a_{i+1/2,j}\frac{U_{i+1,j} - U_{i,j}}{h}.$$

Taking the discrete divergence gives

$$(A_h U)_P = \frac{1}{h^2} \sum_{s=E,W,N,S} a_s (U_P - U_s).$$

- Smooth coefficient: $a_{i+1/2,j} = (a_{i,j} + a_{i+1,j})/2$.
- Share the same $a_s$ in the two adjacent rows.
- For a boundary neighbor, $U_s = g_s$: add $a_s g_s/h^2$ to the right-hand side.

Positive shared weights and boundary elimination yield a symmetric positive-definite system.

# FEM: coefficients and boundary assembly

For test functions $v_h = 0$ on the boundary,

$$\int_\Omega a \nabla u_h \cdot \nabla v_h = \int_\Omega f v_h.$$

Write $u_h = \sum_j U_j \phi_j$ and assemble element contributions:

$$K_{ij} = \sum_T \int_T a \nabla \phi_j \cdot \nabla \phi_i, \qquad F_i = \sum_T \int_T f \phi_i.$$

Separate interior and boundary degrees of freedom:

$$K_{II} U_I = F_I - K_{IB} g_B.$$

- Smooth $a$: quadrature integrates coefficient and load terms.
- Layered $a$: fitted elements lie within one material each.

Positive weak energy gives SPD after Dirichlet elimination.

# How we measure error

**Average error over the domain:**

$$E_{L^2} = \left( \int_\Omega |u^* - u_h|^2 \, dx \right)^{1/2}.$$

**Largest error at the sampled nodes:**

$$E_{\infty,\text{nodes}} = \max_i |u^*(x_i) - U_i|.$$

FDM: integrate a reconstructed function; FEM: integrate the P1 function. The nodal maximum is a discrete $L^\infty$ measure, not the continuous supremum.

For successive mesh sizes $h, \widetilde{h}$,

$$p = \frac{\log(E_h / E_{\widetilde{h}})}{\log(h/\widetilde{h})}.$$

# Why smooth FDM is second order

At a face midpoint $m$,

$$\frac{u(x+h)-u(x)}{h} = u'(m) + \frac{h^2}{24}u'''(m) + O(h^4).$$

The coefficient average also approximates $a(m)$ to order $h^2$. Taking the flux difference gives the consistency defect

$$\tau_h = b_h - A_h R_h u^* = O(h^2).$$

The nodal error satisfies

$$A_h(U_h - R_h u^*) = \tau_h.$$

**Consistency + stability** gives second-order solution error for this smooth, uniformly elliptic problem.

# Why P1 FEM is second order in $L^2$

Let $B(w,v) = \int_\Omega a\nabla w \cdot \nabla v$ and $e = u - u_h$. Galerkin orthogonality gives $B(e, v_h) = 0$.

$$\|e\|_a \le \|u - I_h u\|_a \le Ch|u|_{H^2}.$$

For the dual problem $B(v,z) = (v,e)$,

$$\|e\|_{L^2}^2 = B(e, z - I_h z) \le Ch\|e\|_a \|z\|_{H^2}.$$

With dual regularity $\|z\|_{H^2} \le C\|e\|_{L^2}$,

$$\boxed{\|e\|_{L^2} \le Ch^2}, \qquad \|e\|_a = O(h).$$

This does not by itself prove an unconditional $O(h^2)$ maximum-norm bound.

# Separating numerical error sources

$$\text{total error} = \text{discretization} + \text{quadrature} + \text{algebraic error}.$$

The smooth benchmark uses

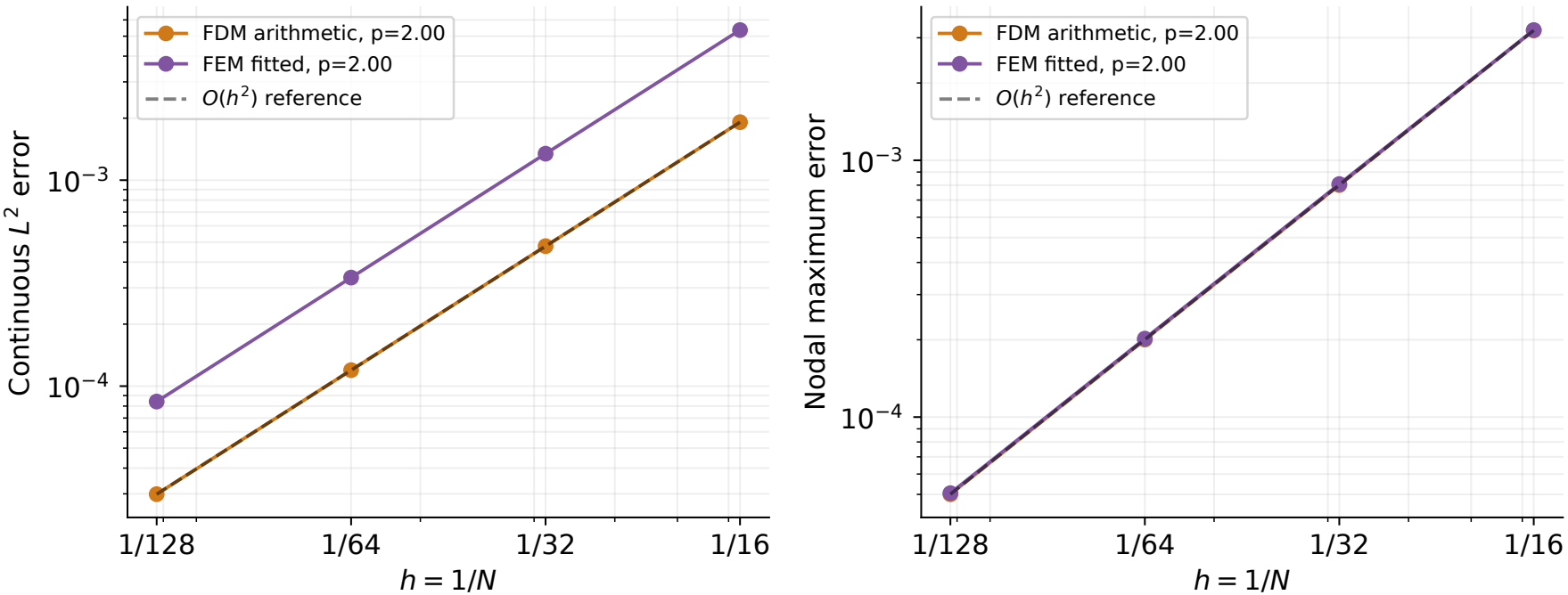
$$u^* = \sin(\pi x)\sin(\pi y), \quad a = 1 + x, \quad g = 0,$$

$$f = 2\pi^2(1 + x)\sin(\pi x)\sin(\pi y) - \pi\cos(\pi x)\sin(\pi y).$$

- Independently check $f$ by differentiating the analytic flux numerically.
- Estimate algebraic error through a residual-correction solve.
- Compare FEM quadrature orders 6 and 8.

Both extra effects are below $10^{-3}$ of the smooth measured $L^2$ error.

# Smooth benchmark: convergence in both measures




Four meshes: $N = 16, 32, 64, 128$, with $h = 1/N$. Left: continuous $L^2$; right: nodal maximum. Dashed black lines indicate slope two; fitted orders use the finest three meshes.

Both methods approach second order here; error constants differ.

# Why the interface needs a different argument

At $x = 1/2$, $a$ and $u_x$ jump. A Taylor expansion through that point is invalid.

The physically useful invariant is instead

$$q_x = -a u_x = \text{constant}.$$

- **FDM**: what coefficient gives the correct resistance across an edge?
- **FEM**: can the approximation space represent the slope change?

The two methods answer these questions differently.

# FDM: derive the interface resistance

Integrate $u_x = -q_x/a$ along an edge crossing the interface:

$$u_R - u_L = -q_x \left( \frac{d_L}{a_L} + \frac{d_R}{a_R} \right).$$

Match this to $q_x = -a_{\text{edge}}(u_R - u_L)/h$, where $h = d_L + d_R$:

$$\boxed{a_{\text{edge}} = \frac{h}{d_L/a_L + d_R/a_R}}.$$

For a bisected edge,

$$a_H = \frac{2a_L a_R}{a_L + a_R} = \frac{20}{11}.$$

The two material resistances add in series.

# Why arithmetic averaging fails here

For the same edge,

$$a_A = \frac{1+10}{2} = 5.5, \qquad a_H = \frac{20}{11} \approx 1.818.$$

Inserting the *exact nodal temperature drop* into each flux formula gives

$$\frac{q_A[u^*]}{q^*} = \frac{a_A}{a_H} = \frac{121}{40}.$$

- Arithmetic averaging assigns too little thermal resistance.
- The computed temperature adjusts when the full system is solved.
- 121/40 is a consistency check, not the ratio of re-solved fluxes.

Positive coefficients can still give symmetry and boundedness while encoding the wrong interface physics.

# FEM: why fit the interface?

On each triangle, P1 FEM has a constant gradient.

- A triangle crossing the interface cannot represent both slopes.
- A fitted mesh places the kink on element edges.
- Integrate $a$ within each material; impose exact boundary data.

For this benchmark, $u^*$ is continuous and affine within each material:

$$u^* \in V_h, \qquad B(u^* - u_h, v_h) = 0.$$

Choosing $v_h = u^* - u_h$ gives

$$\|u^* - u_h\|_a^2 = 0.$$

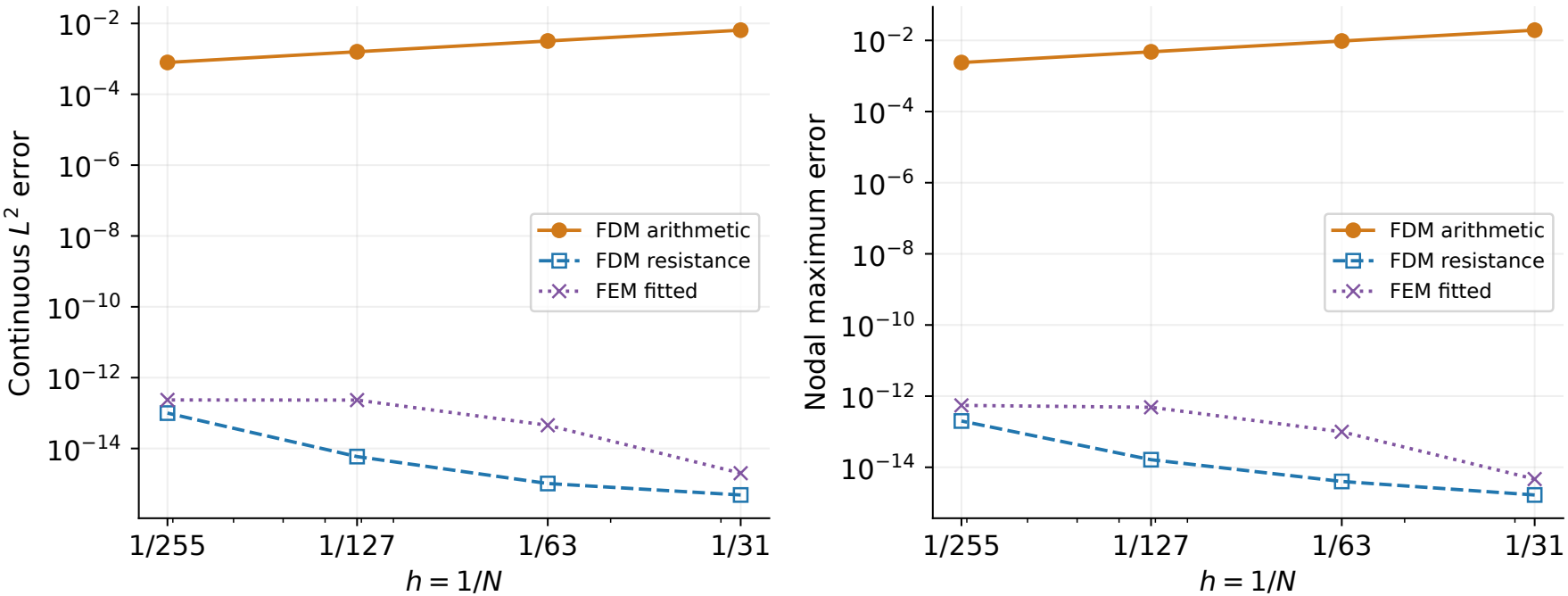
Hence fitted FEM reproduces this solution in exact arithmetic.

# Three methods in the layered comparison

| Method | Interface treatment | Plot identity |
|---|---|---|
| FDM arithmetic | Arithmetic face coefficient | Orange circles; solid line |
| FDM resistance | Subedge resistance; harmonic for equal lengths | Blue hollow squares; dashed line |
| FEM fitted | Interface on triangle edges; material-wise integration | Purple crosses; dotted line |

FDM uses odd $N = 31, 63, 127, 255$; FEM inserts the interface column. Macro spacing is comparable, but unknown counts differ.

# Layered refinement: $L^2$ and nodal maximum error




As for the smooth model: continuous $L^2$ error on the left, nodal maximum error on the right, with $h = 1/N$ on four meshes.

Arithmetic errors decrease; the other two methods already have tiny errors that do not follow the smooth model's convergence pattern. Why?

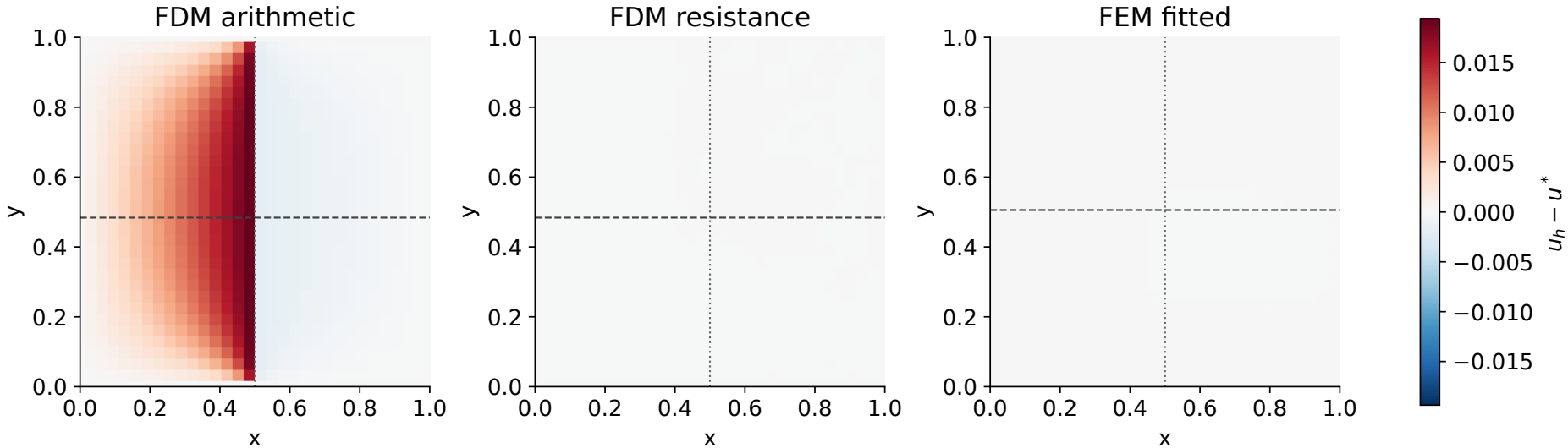
# Why the usual convergence test is inconclusive here

This benchmark is **exactly representable**, rather than simply very smooth:

- $u^*$ is affine within each layer, with a kink at the interface.
- Resistance FDM gives the exact nodal solution; fitted P1 FEM contains $u^*$.
- Their discretization error is zero in exact arithmetic. Floating-point computation and the linear solve leave small measured errors.

We cannot infer a general convergence order from those tiny errors. The benchmark instead tests whether the interface is represented correctly.

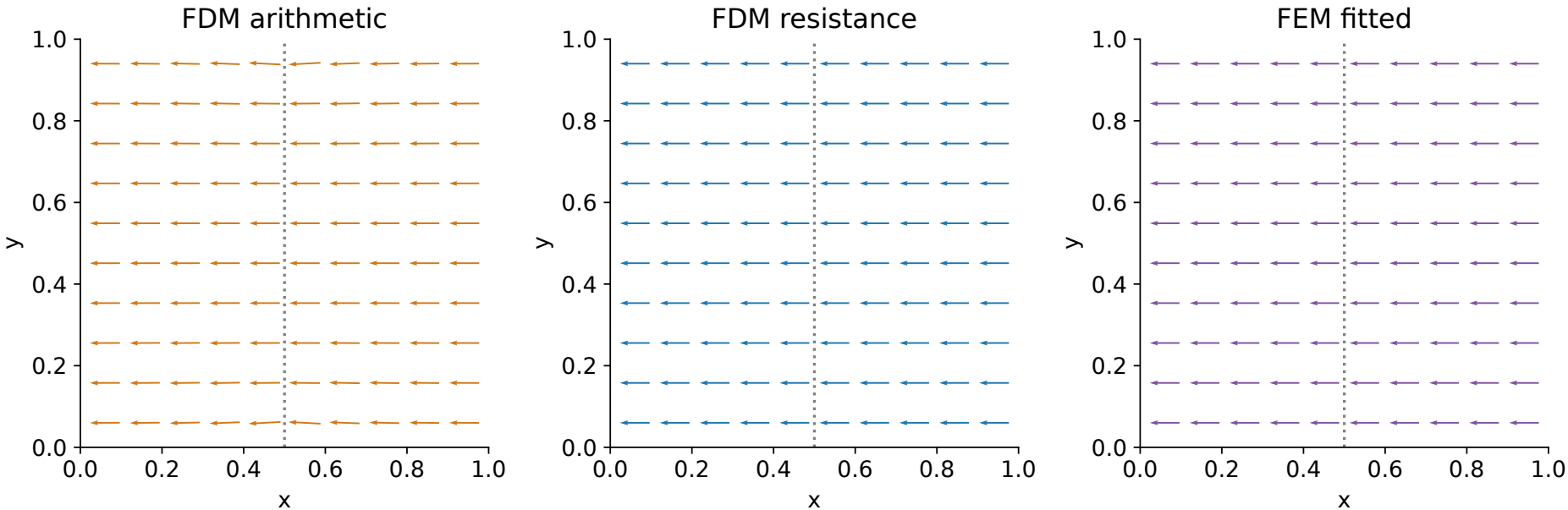
# Where does the layered temperature error occur?




Signed nodal error $u_h - u^*$ at $N = 31$, with one shared color scale. Vertical dotted lines mark the interface; horizontal dashed lines mark the slices used below.

Arithmetic error concentrates around the interface and varies with $y$. Resistance FDM and fitted FEM are essentially zero on this scale.

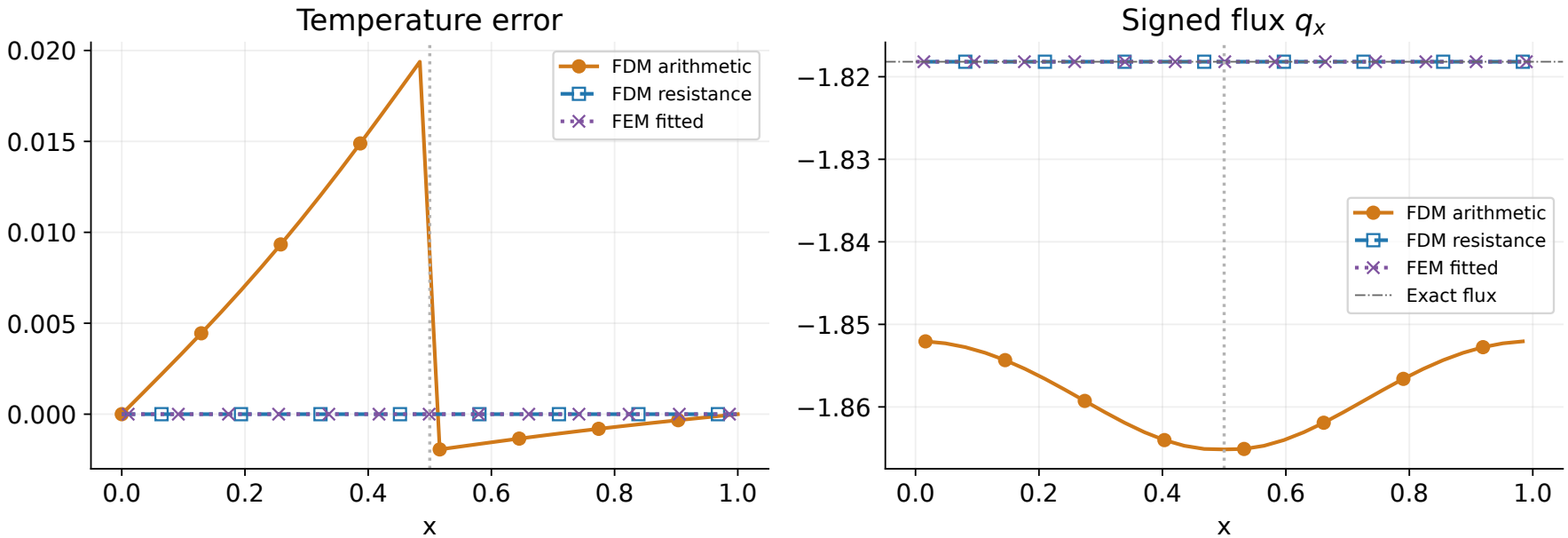
# Inspect the flux field before taking a slice




Arrows show numerical $\mathbf{q}_h$, enlarged with the same scale for all methods; lengths are not normalized. Dotted lines mark $x = 1/2$.

The exact field is $\mathbf{q}^* = (-20/11, 0)$: horizontal and independent of $y$. Arithmetic FDM introduces small changes in both magnitude and direction.

# Horizontal slices reveal the signed flux error




Along $y = y_0$, plot $u_h - u^*$ and the interface-normal component $q_{x,h}(x, y_0)$. FDM: $y_0 = 0.484$; FEM: $y_0 = 0.506$; $N = 31$. The gray reference is $q_x^* = -20/11$; the dotted vertical line is the interface.

This slice checks normal flux; it does not replace the full 2D field. Blue squares and purple crosses recover the reference. Arithmetic FDM does not.

# What validates the interface result?

- Compare signed flux with the exact $-20/11$.
- Reconstruct fluxes independently on the two material sides; sharing one face flux is not an independent continuity check.
- All layered solutions remain between boundary extrema 0 and 1.
- Positive diagonal and nonpositive off-diagonal entries support a discrete maximum principle on these meshes.

Bounded temperature does *not* prove accuracy: arithmetic FDM passes that check too.

# Refinement and contrast increase solver work

Unpreconditioned CG: refining $N = 15 \to 63$

| Contrast | Resistance FDM | Fitted FEM |
|---:|---:|---:|
| 1 | $44 \to 197$ | $53 \to 221$ |
| 10 | $90 \to 470$ | $94 \to 472$ |
| 100 | $168 \to 1190$ | $189 \to 1205$ |

Entries are iteration counts, with zero initial guess and $\|b - AU\|_2 / \|b\|_2 \leq 10^{-10}$ in every case.

Counts show increasing difficulty in the tested cases; they are not condition-number estimates or wall-time rankings.

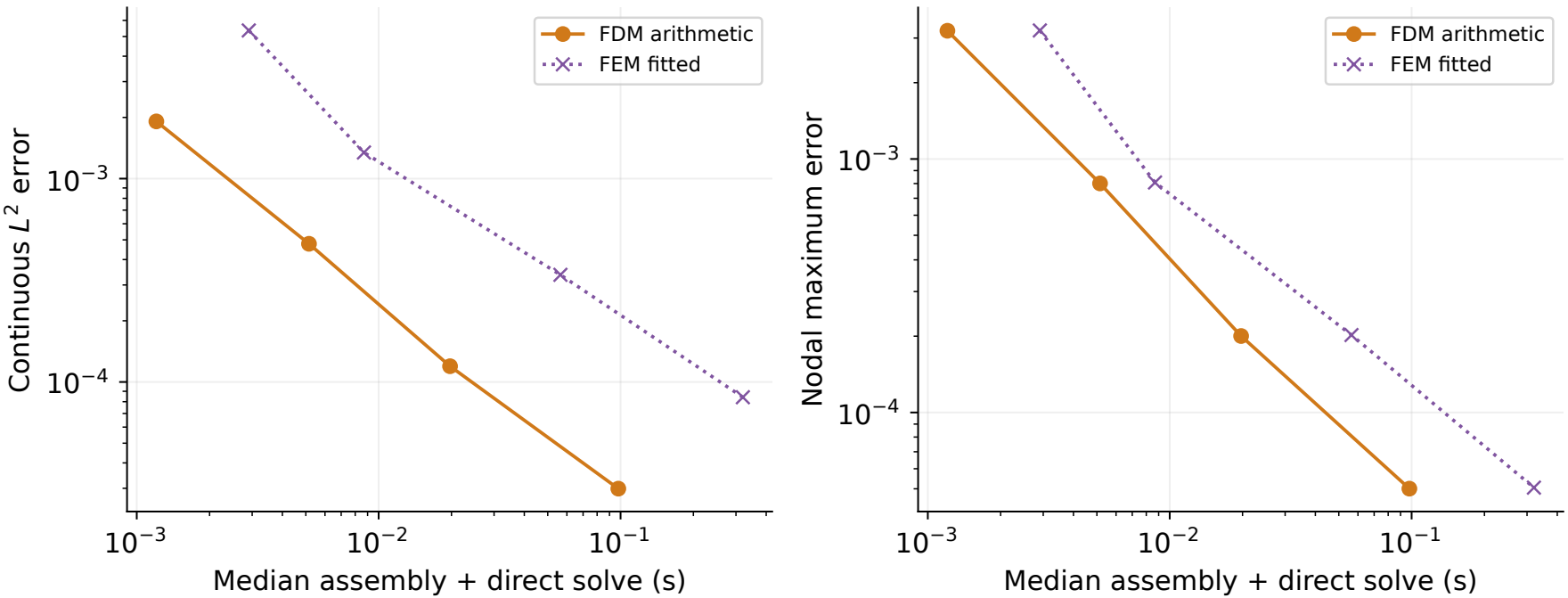
# Cost versus accuracy: what is measured?

$$T = T_{\text{assembly}} + T_{\text{direct solve}}.$$

- One warm-up, then three timed repetitions per case; plot the median.
- Exclude error diagnostics, plotting and file I/O from the timing.
- Use verified errors and the same mesh cases as the refinement studies.
- Compare these implementations; FEM and FDM assembly costs differ.

In 2D, unknown count and sparse assembly/storage scale as $O(h^{-2})$. Sparse direct factorization can scale as $O(h^{-3})$ with suitable ordering; actual timings depend on implementation and hardware.

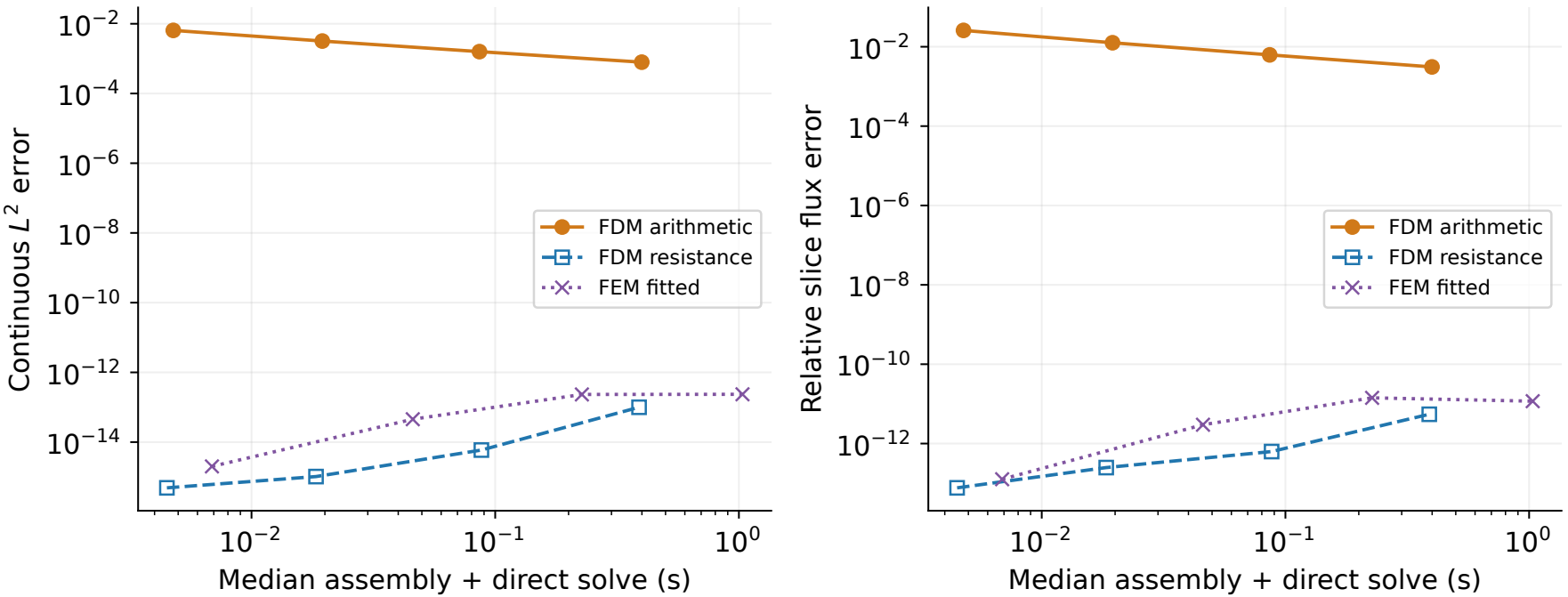
# Smooth model: cost versus accuracy




Horizontal axis: median assembly plus direct-solve time in seconds. Vertical axes: continuous $L^2$ and nodal maximum errors. Each marker is one mesh; connecting lines follow refinement.

FDM attains lower errors at lower measured cost in these runs. This describes these implementations and this benchmark.

# Layered model: cost versus accuracy




Left: continuous $L^2$ error. Right: maximum relative signed-flux error along the sampled slice, $\max |q_{x,h} - q_x^*| / |q_x^*|$. Time includes assembly plus direct solve, using the same timing protocol.

Refinement reduces the arithmetic error. Exact representations already reach numerical precision; added work does not give a comparable reduction.

# Conclusions

1. Both methods approach second-order smooth $L^2$ accuracy.
2. Correct interface resistance or a fitted approximation space reproduces this layered benchmark; symmetry alone is insufficient.
3. Flux diagnostics expose errors that temperature plots can hide.
4. Refinement and contrast increase work; accuracy gains depend on what error remains to be reduced.

# Coverage and remaining scope

| Problem-pack task | Evidence and presentation emphasis |
|---|---|
| Assembly, symmetry, SPD | Derivations shown; matrix/eigenvalue diagnostics saved |
| Smooth errors and verification | Both norms shown; full residual tables and forcing checks summarized |
| Interface and normal flux | Central comparison; side-flux mismatch values in saved tables |
| Boundary extrema | Checked and explained through matrix signs |
| Refinement and contrast | Iteration comparison shown; cost study added |

**Not studied:** unresolved FEM, curved interfaces, contact resistance, AMG comparisons, or rigorous floating-point error bounds.